Function-al vectors

Some fourier analysis and linear algebra

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Definition 1.1 Vector Space

Let $\mathbb{F}$ be a field. A **vector space over** $\mathbb{F}$ is a non-empty set V, together with two operations:

1. **Vector addition:**
$$+ : V \times V \to V, \qquad (u, v) \mapsto u + v.$$
2. **Scalar multiplication:**
$$\cdot : \mathbb{F} \times V \to V, \qquad (a, v) \mapsto av.$$

These operations must satisfy the following axioms for all $u, v, w \in V$ and $a, b \in \mathbb{F}$:

A. Axioms of vector addition

1. **Commutativity:** $u + v = v + u$.
2. **Associativity:** $(u + v) + w = u + (v + w)$.
3. **Additive identity:** There exists $0_V \in V$ such that $v + 0_V = v$.
4. **Additive inverse:** For every $v \in V$, there exists $-v \in V$ such that $v + (-v) = 0_V$.

B. Axioms of scalar multiplication

5. **Distributivity over vector addition:** $a(u + v) = au + av$.
6. **Distributivity over scalar addition:** $(a + b)v = av + bv$.
7. **Associativity of scalar multiplication:** $a(bv) = (ab)v$.
8. **Scalar identity:** $1_{\mathbb{F}} v = v$, where $1_{\mathbb{F}}$ is the multiplicative identity of the field.

The elements of V are called **vectors**, and the elements of $\mathbb{F}$ are called **scalars**.